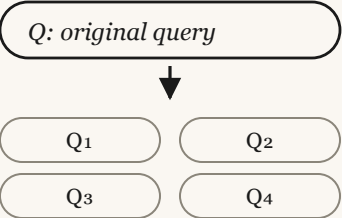


ReformIR — picking reformulations per query

Same bandit machinery as ORE — but the features are reformulations, not signals

REFORMULATOR

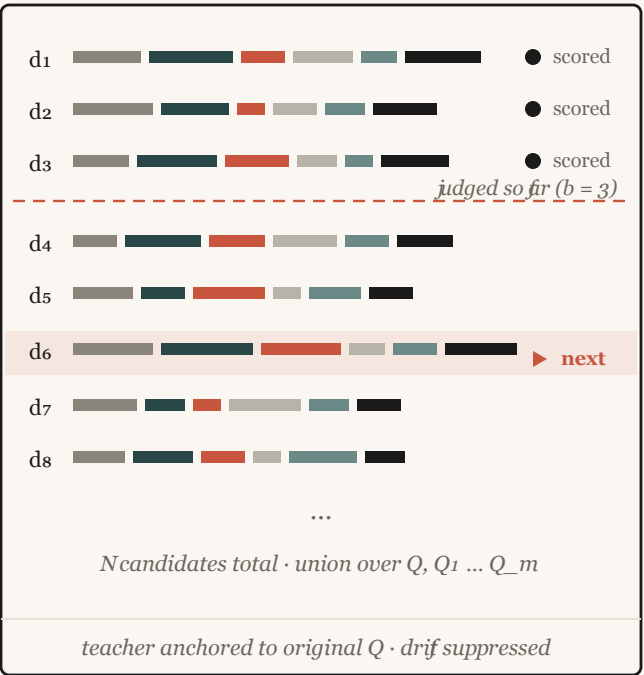
small LLM · cheap



CANDIDATE POOL

each doc carries a feature vector over reformulations

■ BM25(Q) ■ Q1 ■ Q2 ■ Q3 ■ Q4 ■ RM3



LINEAR BANDIT

$$\text{score}(d) = w^\top \psi(d)$$

WEIGHTS w

w_Q		0.36
w_Q1		0.28
w_Q2		0.21
w_Q3		0.01
w_Q4		0.07
w_RM3		0.05